Because the criteria for caching a preflight request are so limited, the preflight result cache provides narrow value. The preflight result cache is only useful for multiple requests to the same URL from the same origin, within a five-minute span. Most requests to an API will probably fall outside of this window. Luckily, there are still changes you can make to your API to reduce preflight requests.

6.6.2 Changing your site to reduce preflight requests

The key to reducing preflight requests is to reduce the conditions that result in preflight requests. Recall from chapter 4, a preflight is issued in cases where:

- The request method is something other than `GET` or `POST`.
- The request contains custom HTTP headers.

Eliminating these conditions will eliminate the need for a preflight. The following section looks at ways in which you can do this. Note that some of these suggestions fly in the face of what is considered good API design. You should weigh your need to eliminate preflight requests against the need to have a sane, structured RESTful API before implementing these suggestions.

Reducing custom headers

If your API includes custom headers on requests, eliminating or moving these headers can remove the need for a preflight. The Content-Type request header indicates what data type is in the body of a request. If you have a JSON API, it's tempting to set the value of the Content-Type request header to `application/json`. In fact, there are some libraries that will automatically set this value for you. But setting the Content-Type to `application/json` triggers a preflight, because the only allowed values for Content-Type are `application/x-www-form-urlencoded`, `multipart/form-data`, or `text/plain`. By removing this header from the request, you avoid the need to issue a preflight.

If you need the information from a particular header, you could consider moving the value from a header to a query parameter. The sample code in chapter 4 introduced the Timezone-Offset and Sample-Source custom headers to demonstrate how preflight requests work. Instead of custom headers, you could use query parameters to convey the same information. Table 6.6 shows two ways to convey the time zone and sample source data introduced in chapter 4.

Table 6.6 Moving custom headers into the query parameter. Eliminating the custom headers eliminates the need for a preflight request.

Original request (requires a preflight request due to custom headers)	New request (no custom headers, so no preflight request)

`GET /api/posts HTTP/1.1`
`Host: 127.0.0.1:9999`
`Origin: http://localhost:1111`
`Timezone-Offset: 300`
`Sample-Source: CORSInAction`
	`GET /api/posts?`
	`  timezoneOffset=300&`
	`  sampleSource=CORSInAction`
	`  HTTP/1.1`
	`Host: 127.0.0.1:9999`
	`Origin: http://localhost:1111`